

Mini project (nosql)

Mini Project Create:

Employee collection

Insert 50 records

Perform:

- 1.Salary analysis
- 2.Department headcount
- 3.Top 5 highest paid employees
- 4.Create index on salary
- 5.Run aggregation pipeline

1.Salary Analysis

1.Avgsalary

```
db.employees.aggregate(  
  [  
    {$group:{  
      _id:null,  
      avgsalary:{$avg:"$salary"}}  
    }  
  ]  
)
```

2.Maxsalary

```
db.employees.aggregate(  
  [  
    {$group:{  
      _id:null,  
      maxsalary:{$max:"$salary"}}  
    }  
  ]  
)
```

3.Minsalary

```
db.employees.aggregate( [  
  {$group:{  
    _id:$departmentl,  
    minsalary:{$min:"$salary"}}  
  }  
)
```

4.Department wise avgsalary

```
db.employees.aggregate(  
  [  
    {$group:{  
      _id:'$department',
```

```

    avgsalary:{$avg:"$salary"}}
  }
]
)

```

5.Department wise maxsalary

```

db.employees.aggregate(
[
  {$group:{
    _id:"$department",
    maxsalary:{$max:"$salary"}}
  }
]
)

```

6.Department wise minsalary

```

db.employees.aggregate(
[
  {$group:{
    _id:"$department",
    minsalary:{$min:"$salary"}}
  }
]
)

```

7.Department wise Headcount

```

db.employees.aggregate([
{
  $group:{
    _id:"$department",
    headcount:{$sum:1}
  }
}
])

```

8.Top 5 High paid Employees

```

db.employees.find().sort({ salary: -1 }).limit(5)

```

Or

```

db.employees.aggregate([
{ $sort: { salary: -1 } },
{ $limit: 1 }
])

```

9.Create index on salary:

```
db.employees.createIndex({ salary: -1 })
```

10.Run aggregation pipeline

```
db.employees.aggregate([
  {
    $group: {
      _id: "$department",
      avgSalary: { $avg: "$salary" },
      maxSalary: { $max: "$salary" },
      minSalary: { $min: "$salary" },
      totalEmployees: { $sum: 1 }
    }
  },
  {
    $sort: { avgSalary: -1 }
  }
])
```